

Engel's Theorem and Lie's Theorem

1.1 Engel's Theorem

Lemma 1.1.1

Suppose that L is a Lie subalgebra of $\mathfrak{gl}(V)$, where V is non-zero, such that every element of L is a nilpotent linear transformation. Then there is some non-zero $v \in V$ such that $xv = 0$ for all $x \in L$.

Proof sketch.

对维数归纳

对于任意 L 的真子代数 A , 模 A 的伴随:

$$\begin{aligned}\varphi : A &\rightarrow \mathfrak{gl}(L/A) \\ a &\mapsto (x + A \mapsto [a, x] + A)\end{aligned}$$

我们看到 $\varphi(A)$ 的公共 0-特征向量, 就是关于 A 代数封闭的向量.

根据归纳假设, $\varphi(A)$ 有非零公共 0-特征向量 y , 从而 $y \notin A$ 且 y 是关于 A 代数封闭的. 这就意味着 A 可以成为一个更大的代数 $\tilde{A} = A \oplus \text{span}\{y\}$ 的理想. 如果我们取 A 是极大的, 那么 $\tilde{A} = L$. 从而 A 成为 L 的一个余维数为 1 的理想. 一个引理是, L 的理想 A 的零子空间 W 是 L 不变的子空间. 这就意味着当我们寻找 $L = A \oplus \text{span}\{y\}$ 的公共零特征向量时, 可以从 A 的零子空间 W 中, 寻找 $\text{span}\{y\}$ 的公共零特征向量, 这是可行的, 因为 $\text{span}\{y\}$ 也是一组幂零的线性变换.

Proof. We induct on the dimension of L . Let A be a maximal subalgebra of L . Define a map

$$\begin{aligned}\varphi : A &\rightarrow \mathfrak{gl}(L/A) \\ a &\mapsto (x + A \mapsto [a, x] + A)\end{aligned}$$

It is straightforward to verify that φ is a Lie algebra homomorphism. Thus $\varphi(A)$ is a Lie algebra. For each $a \in A$, since a is nilpotent, $\text{ad } a$ is as well, hence $\varphi(a)$ is nilpotent. We know from inductive hypothesis that there exists nonzero $y + A \in L/A$ such that $0 = \varphi(a)(y + A)$, $\forall a \in A$. That is $y \notin A$ and

$[A, y] \subseteq A$. By defining $\tilde{A} := A \oplus \text{span}\{y\}$, we know A is an ideal of $\tilde{A}$. Since A is maximal, there must be $\tilde{A} = L$, A is an ideal of L with codimension 1.

A lemma gives that the space W

$$W = \{v \in V : a(v) = 0, \forall a \in A\}$$

is L -invariant. Specifically, $y(W) \subseteq W$. $y|_W$ is nilpotent, there exist nonzero $v \in W$ such that $y(v) = y|_W(v) = 0$. Then for each element of L , we can write it as $a + ky$ for some $k \in F$. Then

$$(a + ky)(v) = a(v) + ky(v) = 0$$

which completes the proof. □

Theorem 1.1.2 ▶ Engel's Theorem

Let V be a vector space. Suppose that L is a Lie subalgebra of $\mathfrak{gl}(V)$ such that every element of L is a nilpotent linear transformation of V . Then there is a basis of V in which every element of L is represented by a strictly upper triangular matrix.

Proof. We induct on the dimension of V . By Lemma ??, there exists $w \in V$ such that $x(w) = 0, \forall x \in L$. Let $W = \text{span}\{w\}$. Then every x induces a linear transformation $\bar{x} : V/W \rightarrow V/W$, $\bar{x}(v + W) = x(v) + W$. It is easily to check that the map

$$L \rightarrow \mathfrak{gl}(V/W), \quad x \mapsto \bar{x}$$

is a Lie homomorphism.

The image of L under this homomorphism is a subalgebra of $\mathfrak{gl}(V/W)$, which satisfies the hypothesis of Engel's Theorem. Moreover $\dim(V/W) = \dim V - 1$. We know from inductive hypothesis that there is a basis of V/W , say $\{v_i + W\}_{2 \leq i \leq n}$ in which every element of L is represented by a strictly upper triangular matrix. That is equivalent to for each $x \in L$,

$$\bar{x}(v_j + W) \in \text{span}\{v_2 + W, \dots, v_{j-1} + W\}, \quad \forall 3 \leq j \leq n,$$

and

$$\bar{x}(v_2 + W) = 0 + W$$

By taking $v_1 = w$, it follows that

$$x(v_1) = 0, \quad x(v_2) \in \text{span}\{v_1\}$$

$$x(v_j) \in \text{span}\{v_1, v_2, \dots, v_{j-1}\}, \quad \forall 3 \leq j \leq n$$

That is every element of L is represented by a strictly upper triangular matrix under the basis $\{v_1, v_2, \dots, v_n\}$. $\square$

Definition 1.1.3

Recall that a Lie algebra is said to be nilpotent if for some $m \geq 1$ we have $L^m = 0$ or, equivalently, if for all $x_0, x_1, \dots, x_m \in V$ we have

$$[x_0, [x_1, \dots, [x_{m-1}, x_m] \dots]] = 0.$$

Theorem 1.1.4 ► Engel's Theorem, second version

A Lie algebra L is nilpotent if and only if for all $x \in L$ the linear map $\text{ad } x : L \rightarrow L$ is nilpotent.

Proof. The 'only if' is straightforward. For the 'if' part, if all $\text{ad } x$ is nilpotent, the algebra

$$\text{ad}(L) = \{\text{ad } x : x \in L\}$$

is a subalgebra of $\mathfrak{gl}(L)$ satisfying the hypothesis of Engel's Theorem. it follows from Engel's Theorem that there exists a basis of L say $\{e_1, \dots, e_n\}$, such that

$$[L, e_i] \subseteq \text{span}\{e_1, \dots, e_{i-1}\}$$

Denote L_j by $\text{span}\{e_1, \dots, e_j\}$, then

$$[L, L_j] \subseteq L_{j-1}, j \geq 2, \quad [L, L_1] = 0$$

We have

$$L^0 = L = L_n$$

$$L^1 = [L, L] \subseteq L_{n-1}$$

Inductively, we have

$$L^k = [L, L^{k-1}] \subseteq [L, L_{n-k+1}] \subseteq L_{n-k}$$

Finally, we have

$$L^n = 0$$

L is nilpotent. □

1.2 Lie's Theorem

Proposition 1.2.1

Let V be a non-zero complex vector space. Suppose that L is a solvable Lie subalgebra of $\mathfrak{gl}(V)$. Then there is some non-zero $v \in V$ which is a simultaneous eigenvector for all $x \in L$.

Proof sketch.

- 一个李代数 L 的子代数 A 如果包含了 L' , 则 A 是 L 的一个理想.
- 如果 L 是可解的, 则 L' 是真子空间, 这意味着我们可以构造 L 的余一维的理想.
- 归纳假设给出 A 的一个特征向量, 从而对应的 weight 子空间非空.
- 理想的 Weight 子空间是 L -不变的, 特别地是 v -不变的.
- 由于空间是 $\mathbb{C}$ 上的, 在 Weight 子空间上存在特征向量 v , 它就是公共的特征向量.

Theorem 1.2.2 ▶ Lie's Theorem

Let V be an n -dimensional complex vector space and let L be a solvable Lie subalgebra of $\mathfrak{gl}(V)$. Then there is a basis of V in which every element of L is represented by an upper triangular matrix.

Proof sketch.

证明完全类似于 Engel 定理的证明.